

Wellness CA -- FastAPI Endpoint Reference

Server: 152.42.181.66:8000 | Auth: JWT Bearer + X-API-Token

This document describes all 8 API endpoints provided by the Wellness CA FastAPI backend. All endpoints (except /register and /login) require a valid JWT token in the Authorization header. All endpoints require the X-API-Token gateway header.

POST /register

Create a new user account. Password is bcrypt-hashed before storage.

Request (JSON):

```
{
  "username": "alice",
  "password": "secret123"
}
```

Response 201:

```
{
  "message": "Registered",
  "user_id": 1
}
```

Errors: 400 (username exists), 403 (invalid gateway token)

POST /login

Authenticate user. Returns a signed JWT access token valid for 24 hours.

Request (JSON):

```
{
  "username": "alice",
  "password": "secret123"
}
```

Response 200:

```
{
  "access_token": "eyJhbGciOi...",
  "token_type": "bearer"
}
```

JWT payload: {"sub": "1", "exp": 1782433600}

Errors: 400 (bad credentials), 403 (gateway)

GET /records

List the current user's wellness records, newest first.

Headers: Authorization: Bearer <token>, X-API-Token: <gateway>

Response 200:

```
[
  {
    "id": 1,
    "sleep_hours": 7.5,
    "exercise_activity": "Running",
    "exercise_duration": 30,
    "record_date": "2025-07-01",
    "notes": "Felt great"
  }
]
```

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Errors: 401 (invalid/expired JWT), 403 (gateway)

POST

/records

Create a new wellness record for the authenticated user.

Request (JSON):

```
{
  "sleep_hours": 7.5,
  "exercise_activity": "Running",
  "exercise_duration": 30,
  "record_date": "2025-07-01",
  "notes": "Felt great"
}
```

Response 201: {"message": "Created", "id": 1}

Errors: 401, 403

PUT

/records/{id}

Update an existing wellness record. User can only edit their own records.

Request body: same shape as POST /records

Response 200: {"message": "Updated"}

Errors: 401, 403, 404 (record not found or not owned)

DELETE

/records/{id}

Delete a wellness record. User can only delete their own.

No request body.

Response 200: {"message": "Deleted"}

Errors: 401, 403, 404

POST

/chat

Send a message to the AI wellness chatbot (DeepSeek). Conversation is saved to database.

Request (JSON):

```
{
  "message": "I've been sleeping poorly this week."
}
```

Response 200:

```
{
  "reply": "Try reducing screen time before bed..."
}
```

Errors: 401, 403

GET

/recommendations

Agentic AI: analyzes the user's last 7 days of wellness data, calls DeepSeek to generate 3 personalized tips, saves them to the database, and returns them.

No request body.

Response 200:

```
{
```

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```
"recommendations": [  
  "Go to bed 30 min earlier",  
  "Add 10 min stretching",  
  "Reduce caffeine after 2pm"  
]  
}
```

Errors: 401, 403. If no records exist, returns a default tip.

Authentication Summary

```
Authorization: Bearer <jwt_access_token>  
X-API-Token: team-wellness-2025  
Content-Type: application/json
```

The JWT is obtained from POST /login. It expires after 24 hours. Android clients should store it in SharedPreferences and redirect to login on 401 responses. The X-API-Token is a shared team secret hardcoded in ApiClient.kt.